THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 13: **Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The albedo of a surface is 0.8. The percentage of sunlight reflected is: 1

(a) 80 per cent **(b)** 0.8 per cent **(c)** 8 per cent **(d)** 20 per cent

2. A car emits 120 g of CO_2 per km. For 50 km it emits: 1

(a) 600 g **(b)** 60 kg **(c)** 6 kg **(d)** 2.4 kg

3. Rainwater percolating into the ground becomes: 1

(a) vapour **(b)** groundwater **(c)** river water **(d)** ice

4. The nitrates in the soil are taken up by: 1

(a) plants through roots **(b)** rocks
(c) air **(d)** animals directly

5. Composting of kitchen waste: 1

(a) reduces methane from landfills **(b)** needs coal
(c) burns waste **(d)** increases pollution

6. The albedo of a forest is 0.1 and that of fresh snow is 0.85. The surface heated more by sunlight is the: 1

(a) snow **(b)** forest **(c)** both equally **(d)** neither

7. A bulb of 60 W is replaced by a 9 W LED. If the bulb is used 5 h a day, the energy saved in 10 days is: 1

(a) 25.5 kWh **(b)** 0.255 kWh **(c)** 3 kWh **(d)** 2.55 kWh

8. Fog is formed by: 1

(a) transpiration **(b)** precipitation
(c) condensation near the ground **(d)** evaporation

9. Animals get nitrogen from: 1

(a) water **(b)** plants and other animals
(c) air **(d)** soil directly

10. Planting a tree in the school helps by: 1

(a) increasing global warming **(b)** reducing oxygen
(c) absorbing CO_2 and providing shade **(d)** spoiling the soil

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SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is the CO2 level in the atmosphere higher in winter than in summer in the northern hemisphere? 2

12. Why do cities become hotter than villages (urban heat island)? 2

13. Why does melting of the Arctic ice cause further warming? 2

14. State two reasons why CO2 is called a long-lived greenhouse gas. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how the temperatures of the Earth are controlled by the balance of incoming and outgoing radiation. 3

16. Why is the troposphere the layer which contains the weather? Explain with two reasons. 3

17. Describe how mountains affect the weather and the water cycle in a region. 3

OR

Calculate the area of a solar panel needed to produce 500 W if the sunlight is 1000 W/m squared and the efficiency is 20 per cent. 3

18. A family uses 300 units of electricity a month. If each unit emits 0.8 kg of CO2, calculate the annual emission and the saving if 20 per cent is saved. 3

OR

Explain why wetlands are important for the Earth system. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the carbon cycle with a flow chart in words for atmosphere, producers, consumers, decomposers, fossil fuels, and the oceans, and mark the human interference. 5

OR

Explain how the biosphere has changed the atmosphere over geological time, with the rise of oxygen from photosynthesis and the formation of the ozone layer. 5

20. Discuss the evidences of global warming and climate change and the actions of the countries under international agreements. 5

OR

Plan a school-level action under Mission LiFE with the activities, the targets and the way to measure the savings. 5

END OF QUESTION PAPER — Total: 40 Marks